

#This code will be used to demonstrate proposition 8.10. We demonstrate the existence of the curve γ_9

#Construct the polynomial g_3 described in the text of section 8.3.

$$\begin{aligned}
 g_3 &:= \left(- (z^{-3} - t^{-3}) \cdot (y - x) - (w^{-3} - t^{-3}) \cdot (y + x) \right) \cdot \left((z^{-3} - s^{-3}) \cdot (y - x) - (w^{-3} - s^{-3}) \cdot (y + x) \right) - 4 \cdot (x^{-3} - t^{-3}) \cdot (y^{-3} - s^{-3}) \cdot x \cdot y \\
 g_3 &:= \left(- \left(\frac{1}{z^3} - \frac{1}{t^3} \right) (y - x) - \left(\frac{1}{w^3} - \frac{1}{t^3} \right) (y + x) \right) \left(\left(\frac{1}{z^3} - \frac{1}{s^3} \right) (y - x) - \left(\frac{1}{w^3} - \frac{1}{s^3} \right) (y + x) \right) - 4 \left(\frac{1}{x^3} - \frac{1}{t^3} \right) \left(\frac{1}{y^3} - \frac{1}{s^3} \right) x y \quad (1)
 \end{aligned}$$

#Describe s and t as a function of the other distances.

$$z := \frac{y - x}{2} :$$

$$w := \frac{y + x}{2} :$$

#Apply the rational parametrization constructed in this section 8.3.

$$\text{parametrization} := \left\{ s = a + \frac{1}{a}, \right.$$

$$t = b + \frac{1}{b},$$

$$y = 2 \cdot \left(a - \frac{1}{a} \right),$$

$$x = 2 \cdot \left(b - \frac{1}{b} \right) \Big\} :$$

$$\text{simplify}(\text{numer}(\text{expand}(\text{subs}(\text{parametrization}, g_3))));$$

$$\text{simplify}(\text{denom}(\text{expand}(\text{subs}(\text{parametrization}, g_3))));$$

$$\begin{aligned}
 &143 b^3 (a + b)^2 (a + 1) (a b - 1)^2 (a - 1) (-b + a)^2 a^4 (b - 1) (b + 1) (a b + 1)^2 \left(\right. \\
 &\quad - \frac{64 a^9 (a - 1)^2 (a + 1)^2 b^{24}}{143} + \frac{15 \left(a^6 - \frac{1}{5} a^4 + 3 a^2 - \frac{1}{15} \right) a^8 b^{23}}{143} \\
 &\quad \left. + \frac{64 a^7 (a^4 + 4 a^2 + 1) (a - 1)^2 (a + 1)^2 b^{22}}{143} \right)
 \end{aligned}$$

$$\begin{aligned}
& - \frac{60 \left(a^{10} - \frac{3}{20} a^8 + \frac{131}{20} a^6 - \frac{7}{60} a^4 + \frac{77}{20} a^2 - \frac{1}{15} \right) a^6 b^{21}}{143} \\
& - \frac{64 a^5 (a^8 + 12 a^6 - 2 a^4 + 4 a^2 - 3) (a-1)^2 (a+1)^2 b^{20}}{143} \\
& + \frac{1}{143} \left(218 \left(a^{14} - \frac{67}{109} a^{12} + \frac{783}{218} a^{10} + \frac{369}{218} a^8 + \frac{2145}{218} a^6 - \frac{385}{218} a^4 + \frac{173}{109} a^2 \right. \right. \\
& \left. \left. + \frac{61}{109} \right) a^4 b^{19} \right) \\
& - \frac{192 (a+1)^2 (a-1)^2 \left(a^{12} - \frac{8}{3} a^{10} - \frac{10}{3} a^8 - \frac{4}{3} a^6 + 6 a^4 - \frac{1}{3} \right) a^3 b^{18}}{143} \\
& - \frac{1}{143} \left(316 \left(a^{18} + \frac{67}{158} a^{16} + \frac{371}{158} a^{14} - \frac{649}{316} a^{12} + \frac{6117}{316} a^{10} + \frac{1173}{316} a^8 + \frac{2791}{316} a^6 \right. \right. \\
& \left. \left. - \frac{445}{158} a^4 + \frac{307}{158} a^2 + \frac{63}{79} \right) a^2 b^{17} \right) \\
& + \frac{128 \left(a^8 + 12 a^6 - \frac{1}{2} a^4 + 4 a^2 - 3 \right) (a+1)^2 (a-1)^2 a^5 b^{16}}{143} + \left(a^{22} + \frac{59}{13} a^{20} \right. \\
& + \frac{911}{143} a^{18} - \frac{2039}{143} a^{16} + \frac{6230}{143} a^{14} + \frac{2194}{143} a^{12} + \frac{11634}{143} a^{10} - \frac{586}{143} a^8 + \frac{1401}{143} a^6 \\
& \left. - \frac{257}{143} a^4 + \frac{889}{143} a^2 + \frac{127}{143} \right) b^{15} \\
& + \frac{576 \left(a^{12} - \frac{8}{3} a^{10} - \frac{35}{9} a^8 - \frac{32}{9} a^6 + \frac{49}{9} a^4 - \frac{1}{3} \right) (a+1)^2 (a-1)^2 a^3 b^{14}}{143} + \left(\right. \\
& - \frac{387}{143} a^{22} - \frac{1033}{143} a^{20} + \frac{497}{143} a^{18} - \frac{185}{11} a^{16} + \frac{1426}{143} a^{14} - \frac{1266}{11} a^{12} - \frac{318}{13} a^{10} \\
& \left. - \frac{8014}{143} a^8 + \frac{2539}{143} a^6 - \frac{1279}{143} a^4 - \frac{633}{143} a^2 - \frac{435}{143} \right) b^{13} + \left(3 a^{22} + \frac{647}{143} a^{20} \right.
\end{aligned}$$

$$\begin{aligned}
& + \frac{1249}{143} a^{18} - \frac{2581}{143} a^{16} + \frac{8050}{143} a^{14} + \frac{3414}{143} a^{12} + \frac{16374}{143} a^{10} - \frac{1390}{143} a^8 + \frac{2363}{143} a^6 \\
& - \frac{527}{143} a^4 + \frac{1047}{143} a^2 + \frac{381}{143} \Big) b^{11} \\
& - \frac{576 \left(a^{12} - \frac{8}{3} a^{10} - \frac{35}{9} a^8 - \frac{32}{9} a^6 + \frac{49}{9} a^4 - \frac{1}{3} \right) (a+1)^2 (a-1)^2 a^3 b^{10}}{143} + \Big(\\
& - \frac{129}{143} a^{22} - \frac{67}{11} a^{20} + \frac{255}{143} a^{18} - \frac{1383}{143} a^{16} + \frac{694}{143} a^{14} - \frac{11598}{143} a^{12} - \frac{166}{11} a^{10} \\
& - \frac{6122}{143} a^8 + \frac{187}{13} a^6 - \frac{83}{13} a^4 - \frac{631}{143} a^2 - \frac{145}{143} \Big) b^9 \\
& - \frac{128 \left(a^8 + 12 a^6 - \frac{1}{2} a^4 + 4 a^2 - 3 \right) (a+1)^2 (a-1)^2 a^5 b^8}{143} + \frac{1}{11} \Big(20 \left(a^{18} \right. \\
& + \frac{301}{130} a^{16} - \frac{451}{130} a^{14} + \frac{2809}{260} a^{12} + \frac{87}{20} a^{10} + \frac{1215}{52} a^8 - \frac{631}{260} a^6 + \frac{73}{26} a^4 + \frac{61}{130} a^2 \\
& + \frac{81}{65} \Big) a^2 b^7 \Big) \\
& + \frac{192 (a+1)^2 (a-1)^2 \left(a^{12} - \frac{8}{3} a^{10} - \frac{10}{3} a^8 - \frac{4}{3} a^6 + 6 a^4 - \frac{1}{3} \right) a^3 b^6}{143} \\
& - \frac{1}{143} \Big(134 \left(a^{14} + \frac{179}{67} a^{12} - \frac{383}{134} a^{10} + \frac{2175}{134} a^8 + \frac{399}{134} a^6 + \frac{785}{134} a^4 - \frac{61}{67} a^2 \right. \\
& + \frac{115}{67} \Big) a^4 b^5 \Big) + \frac{64 a^5 (a^8 + 12 a^6 - 2 a^4 + 4 a^2 - 3) (a-1)^2 (a+1)^2 b^4}{143} \\
& + \frac{4 \left(a^{10} + \frac{249}{4} a^8 + \frac{7}{4} a^6 + \frac{407}{4} a^4 + \frac{9}{4} a^2 + 17 \right) a^6 b^3}{143} \\
& - \frac{64 a^7 (a^4 + 4 a^2 + 1) (a-1)^2 (a+1)^2 b^2}{143} - \frac{a^8 (a^6 + 51 a^4 + 3 a^2 + 17) b}{143}
\end{aligned}$$

$$+ \frac{64 a^9 (a-1)^2 (a+1)^2}{143} \Bigg)$$

$$4 (a b + 1)^6 (-b + a)^6 (a^2 + 1)^3 b^2 a^2 (b^2 + 1)^3 (a + b)^6 (a b - 1)^6 (b - 1)^3 (b + 1)^3 (a - 1)^3 (a + 1)^3 \quad (2)$$

#Define r9

$$\begin{aligned} r9 := & \left(-\frac{64 a^9 (a-1)^2 (a+1)^2 b^{24}}{143} + \frac{15 \left(a^6 - \frac{1}{5} a^4 + 3 a^2 - \frac{1}{15} \right) a^8 b^{23}}{143} \right. \\ & + \frac{64 a^7 (a^4 + 4 a^2 + 1) (a-1)^2 (a+1)^2 b^{22}}{143} \\ & - \frac{60 \left(a^{10} - \frac{3}{20} a^8 + \frac{131}{20} a^6 - \frac{7}{60} a^4 + \frac{77}{20} a^2 - \frac{1}{15} \right) a^6 b^{21}}{143} \\ & - \frac{64 a^5 (a^8 + 12 a^6 - 2 a^4 + 4 a^2 - 3) (a-1)^2 (a+1)^2 b^{20}}{143} \\ & + \frac{1}{143} \left(218 \left(a^{14} - \frac{67}{109} a^{12} + \frac{783}{218} a^{10} + \frac{369}{218} a^8 + \frac{2145}{218} a^6 - \frac{385}{218} a^4 + \frac{173}{109} a^2 \right. \right. \\ & \left. \left. + \frac{61}{109} \right) a^4 b^{19} \right) \\ & - \frac{192 (a+1)^2 (a-1)^2 \left(a^{12} - \frac{8}{3} a^{10} - \frac{10}{3} a^8 - \frac{4}{3} a^6 + 6 a^4 - \frac{1}{3} \right) a^3 b^{18}}{143} \\ & - \frac{1}{143} \left(316 \left(a^{18} + \frac{67}{158} a^{16} + \frac{371}{158} a^{14} - \frac{649}{316} a^{12} + \frac{6117}{316} a^{10} + \frac{1173}{316} a^8 + \frac{2791}{316} a^6 \right. \right. \\ & \left. \left. - \frac{445}{158} a^4 + \frac{307}{158} a^2 + \frac{63}{79} \right) a^2 b^{17} \right) \\ & + \frac{128 \left(a^8 + 12 a^6 - \frac{1}{2} a^4 + 4 a^2 - 3 \right) (a+1)^2 (a-1)^2 a^5 b^{16}}{143} + \left(a^{22} + \frac{59}{13} a^{20} \right. \\ & + \frac{911}{143} a^{18} - \frac{2039}{143} a^{16} + \frac{6230}{143} a^{14} + \frac{2194}{143} a^{12} + \frac{11634}{143} a^{10} - \frac{586}{143} a^8 + \frac{1401}{143} a^6 \\ & \left. - \frac{257}{143} a^4 + \frac{889}{143} a^2 + \frac{127}{143} \right) b^{15} \end{aligned}$$

$$\begin{aligned}
& + \frac{576 \left(a^{12} - \frac{8}{3} a^{10} - \frac{35}{9} a^8 - \frac{32}{9} a^6 + \frac{49}{9} a^4 - \frac{1}{3} \right) (a+1)^2 (a-1)^2 a^3 b^{14}}{143} + \left(\right. \\
& - \frac{387}{143} a^{22} - \frac{1033}{143} a^{20} + \frac{497}{143} a^{18} - \frac{185}{11} a^{16} + \frac{1426}{143} a^{14} - \frac{1266}{11} a^{12} - \frac{318}{13} a^{10} \\
& - \frac{8014}{143} a^8 + \frac{2539}{143} a^6 - \frac{1279}{143} a^4 - \frac{633}{143} a^2 - \frac{435}{143} \left. \right) b^{13} + \left(3 a^{22} + \frac{647}{143} a^{20} + \frac{1249}{143} a^{18} \right. \\
& - \frac{2581}{143} a^{16} + \frac{8050}{143} a^{14} + \frac{3414}{143} a^{12} + \frac{16374}{143} a^{10} - \frac{1390}{143} a^8 + \frac{2363}{143} a^6 - \frac{527}{143} a^4 \\
& + \frac{1047}{143} a^2 + \frac{381}{143} \left. \right) b^{11} \\
& - \frac{576 \left(a^{12} - \frac{8}{3} a^{10} - \frac{35}{9} a^8 - \frac{32}{9} a^6 + \frac{49}{9} a^4 - \frac{1}{3} \right) (a+1)^2 (a-1)^2 a^3 b^{10}}{143} + \left(\right. \\
& - \frac{129}{143} a^{22} - \frac{67}{11} a^{20} + \frac{255}{143} a^{18} - \frac{1383}{143} a^{16} + \frac{694}{143} a^{14} - \frac{11598}{143} a^{12} - \frac{166}{11} a^{10} - \frac{6122}{143} a^8 \\
& + \frac{187}{13} a^6 - \frac{83}{13} a^4 - \frac{631}{143} a^2 - \frac{145}{143} \left. \right) b^9 \\
& - \frac{128 \left(a^8 + 12 a^6 - \frac{1}{2} a^4 + 4 a^2 - 3 \right) (a+1)^2 (a-1)^2 a^5 b^8}{143} + \frac{1}{11} \left(20 \left(a^{18} \right. \right. \\
& + \frac{301}{130} a^{16} - \frac{451}{130} a^{14} + \frac{2809}{260} a^{12} + \frac{87}{20} a^{10} + \frac{1215}{52} a^8 - \frac{631}{260} a^6 + \frac{73}{26} a^4 + \frac{61}{130} a^2 \\
& + \frac{81}{65} \left. \right) a^2 b^7 \left. \right) + \frac{192 (a+1)^2 (a-1)^2 \left(a^{12} - \frac{8}{3} a^{10} - \frac{10}{3} a^8 - \frac{4}{3} a^6 + 6 a^4 - \frac{1}{3} \right) a^3 b^6}{143} \\
& - \frac{134 \left(a^{14} + \frac{179}{67} a^{12} - \frac{383}{134} a^{10} + \frac{2175}{134} a^8 + \frac{399}{134} a^6 + \frac{785}{134} a^4 - \frac{61}{67} a^2 + \frac{115}{67} \right) a^4 b^5}{143} \\
& + \frac{64 a^5 (a^8 + 12 a^6 - 2 a^4 + 4 a^2 - 3) (a-1)^2 (a+1)^2 b^4}{143} \\
& + \frac{4 \left(a^{10} + \frac{249}{4} a^8 + \frac{7}{4} a^6 + \frac{407}{4} a^4 + \frac{9}{4} a^2 + 17 \right) a^6 b^3}{143}
\end{aligned}$$

$$- \frac{64 a^7 (a^4 + 4 a^2 + 1) (a - 1)^2 (a + 1)^2 b^2}{143} - \frac{a^8 (a^6 + 51 a^4 + 3 a^2 + 17) b}{143} + \frac{64 a^9 (a - 1)^2 (a + 1)^2}{143} \Bigg) :$$

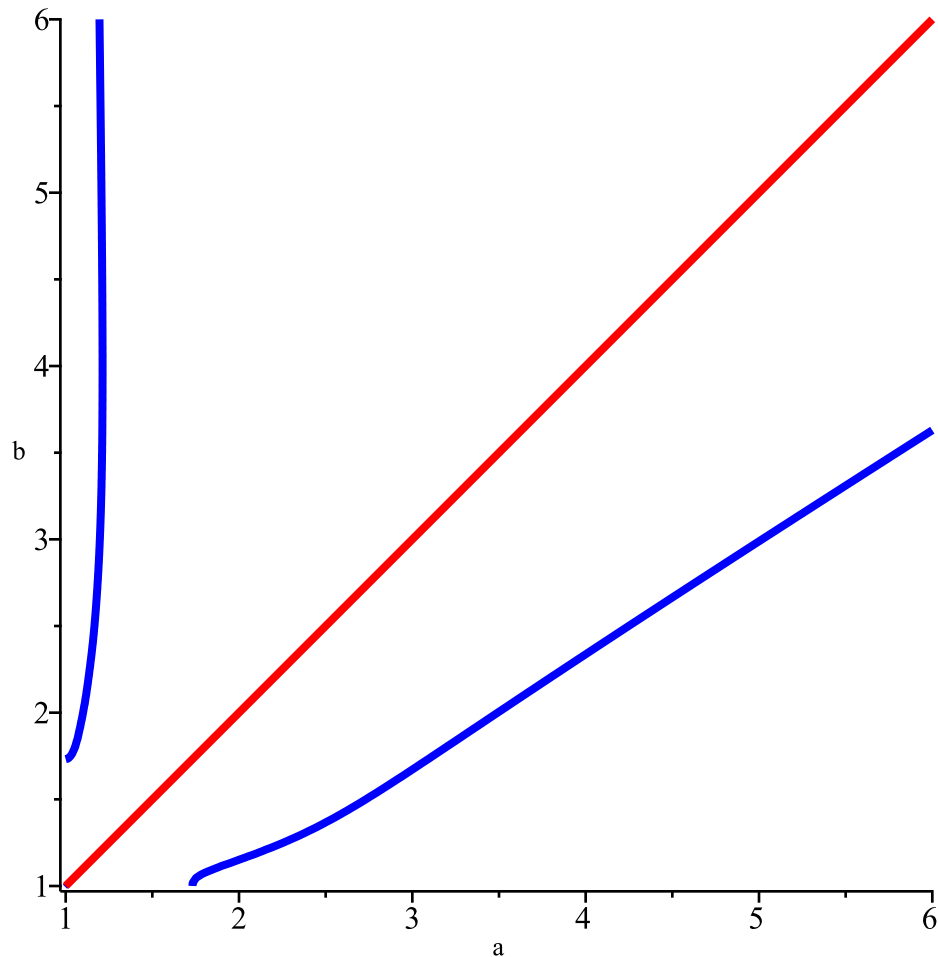
#Plot the curves r9=0 in region R2.

$r := a - b :$

with(plots) :

```
implicitplot([ r9=0, r=0 ], a = 1 .. 6, b = 1 .. 6,
  scaling = constrained,
  gridrefine = 2,
  thickness = 3,
  color = [ blue, red ],
  labels = [ "a", "b" ],
  title = "Graphic for r8=0.");
```

Graphic for r8=0.



$\text{simplify}(\text{eval}(r_9, b = 1))$

$$\frac{8 (7 a^4 - 6 a^2 + 3) (a + 1)^8 (a^2 - 3) (a - 1)^8}{143} \quad (3)$$

#Apply the transformation on parameter region constructing r_9

$\text{transformation} := \{a = c + d + 1, b = d + 1\} :$

$r_9 := \text{collect}(\text{numer}(\text{expand}(\text{subs}(\text{transformation}, r_9))), c) :$

#Compute the degree of r_9 in c

$\text{degree}(r_9, c);$

22

(4)

#Define the coefficients $c_i = \text{coeff}(r_9, c, i)$

$\text{seq}(\text{evalf}(\text{assign}(\text{cat}('c', i), \text{coeff}(r_9, c, i))), i = 0 .. 22);$

#Compute the all degree of c_i in d

$\text{seq}(\text{degree}(\text{eval}(\text{cat}('c', i))), d, i = 0 .. 22);$

37, 36, 35, 34, 33, 32, 31, 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, 18, 17, 16, 15

(5)

#From here we will analyze the signal variation of all coefficients c_i

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{22}, d, i))), i = 15 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(6)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{21}, d, i))), i = 16 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(7)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{20}, d, i))), i = 17 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(8)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{19}, d, i))), i = 18 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(9)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{18}, d, i))), i = 19 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(10)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{17}, d, i))), i = 20 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(11)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{16}, d, i))), i = 21 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(12)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{15}, d, i))), i = 22 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(13)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{14}, d, i))), i = 23 .. 0, -1);$

1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1

(14)

$\text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c_{13}, d, i))), i = 24 .. 0, -1);$

$$\begin{aligned} & \text{seq}(\text{sign}(\text{evalf}(\text{coeff}(c12, d, i))), i = 25 .. 0, -1); \\ & \text{fsolve}(c12=0, d, 0 .. \text{infinity}); \end{aligned} \tag{15}$$

$$\begin{aligned} & \text{1, } -1, -1 \\ & \text{0.04224763972} \end{aligned} \quad (16)$$

$$\begin{aligned} &1, -1, -1, -1, -1 \\ &0.1552381624 \end{aligned} \quad (17)$$

[illegible]

$$\begin{aligned} &1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, \\ &\quad -1, -1, -1 \\ &\qquad\qquad\qquad 1.373948636 \qquad\qquad\qquad (19) \\ &seq(sign(evalf(coeff(c8, d, i))), i=29 \dots 0, -1): \end{aligned}$$

$$-1, \quad (20)$$

$$-1, -1, -1, -1, -1, -1, -1, -1, -1, -1$$

$$\begin{aligned} & seq(sign(evalf(coeff(c7, d, i))), i = 30 \dots 0, -1); \\ & coeff(c7, d, 0) \\ & -1, \\ & \quad -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1 \\ & \qquad \qquad \qquad 0 \end{aligned} \tag{21}$$

```
seq(sign(evalf(coeff(c6, d, i))), i = 31 .. 0, -1);
coeff(c6, d, 1); coeff(c6, d, 0)
-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,
-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1
```

$$\begin{aligned} & 0 \\ & 0 \end{aligned} \tag{22}$$

```
seq(sign(evalf(coeff(c5, d, i))), i = 32 .. 0, -1);
coeff(c5, d, 0); coeff(c5, d, 1); coeff(c5, d, 2)
-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,
-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1
```

$$\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad (23)$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c4, d, i))), i = 33 \dots 0, -1); \\
& coeff(c4, d, 0); coeff(c4, d, 1); coeff(c4, d, 2); coeff(c4, d, 3) \\
& -1, \\
& -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1 \\
& 0 \\
& 0 \\
& 0 \\
& 0 \tag{24}
\end{aligned}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c3, d, i))), i = 34 \dots 0, -1); \\
& coeff(c3, d, 0); coeff(c3, d, 1); coeff(c3, d, 2); coeff(c3, d, 3); coeff(c3, d, 4) \\
& -1, \\
& -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1 \\
& 0 \\
& 0 \\
& 0 \\
& 0 \\
& 0 \tag{25}
\end{aligned}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c2, d, i))), i = 35 \dots 0, -1); \\
& coeff(c2, d, 0); coeff(c2, d, 1); coeff(c2, d, 2); coeff(c2, d, 3); coeff(c2, d, 4); coeff(c2, d, 5) \\
& -1, \\
& -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1 \\
& 0 \\
& 0 \\
& 0 \\
& 0 \\
& 0 \\
& 0 \tag{26}
\end{aligned}$$

$$\begin{aligned}
& seq(sign(evalf(coeff(c1, d, i))), i = 36 \dots 0, -1); \\
& coeff(c1, d, 0); coeff(c1, d, 1); coeff(c1, d, 2); coeff(c1, d, 3); coeff(c1, d, 4); coeff(c1, d, 5); \\
& coeff(c1, d, 6); \\
& -1, \\
& -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1 \\
& 0 \\
& 0 \\
& 0 \\
& 0 \\
& 0
\end{aligned}$$

0

0

(27)

seq(sign(evalf(coeff(c0, d, i))), i = 37 .. 0, -1);

coeff(c0, d, 0) ; coeff(c0, d, 1); coeff(c0, d, 2); coeff(c0, d, 3) ; coeff(c0, d, 4); coeff(c0, d, 5);

coeff(c0, d, 6); coeff(c0, d, 7);

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1,

-1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, 1, 1, 1, 1, 1, 1, 1, 1

0

0

0

0

0

0

0

0

(28)

#This concludes the existence of the curve γ_9 in R_2 .